

Chapter 10

2D problems on the complex plane

10.1 Holomorphic functions

In two dimensions, electrostatics can be mapped onto complex analysis. A *holomorphic* function $\Omega(z)$ is a function of

$$z = x + i y \quad (10.1)$$

alone, *i.e.*

$$\frac{\partial \Omega}{\partial z^*} = 0 \quad (10.2)$$

Writing $\Omega = u + i v$, this condition reads

$$\left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right) (u + i v) = 0 \quad (10.3)$$

which gives the Cauchy–Riemann conditions

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad , \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (10.4)$$

If $\Omega(z)$ is analytic inside a closed contour, then

$$\oint \Omega(z) dz = 0 \quad , \quad \oint \frac{\Omega(\xi)}{\xi - z} d\xi = 2\pi i \Omega(z) \quad (10.5)$$

Otherwise,

$$\oint \Omega(z) dz = 2\pi i \sum \text{residues} \quad (10.6)$$

where, expanding Ω in a Laurent series about a pole at z_0

$$\Omega(z) = \frac{a_{-n}}{(z - z_0)^n} + \cdots + \frac{a_{-1}}{z - z_0} + a_0 + \cdots \quad (10.7)$$

the coefficient a_{-1} is the residue.

For holomorphic functions, differentiating (10.4) gives

$$\frac{\partial^2 u}{\partial x^2} = -\frac{\partial^2 u}{\partial y^2} \Rightarrow \nabla^2 u = 0 \quad , \quad \text{and also} \quad \nabla^2 v = 0 \quad (10.8)$$

that is, u and v solve the Laplace equation in two dimensions. Hence, any $\Omega(z)$ that satisfies the boundary conditions is *the* solution to the corresponding electrostatic problem.

Say $u = \text{Re } \Omega$ is our potential ϕ . Then

$$\frac{d\Omega}{dz} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y} = -E_x + i E_y \quad (10.9)$$

One can also show that

$$\nabla u \cdot \nabla v = 0 \quad (10.10)$$

so that while u plays the role of ϕ , the level curves of v are streamlines perpendicular to the equipotentials – *i.e.* parallel to the electric field lines.

10.2 The complex Laplace problem

The general two-dimensional problem is to solve

$$\nabla^2 \phi = 0 \quad \text{in 2D} \quad (10.11)$$

with boundary condition

$$A(t) \phi + B(t) \frac{\partial \phi}{\partial n} = C(t) \quad (10.12)$$

on a boundary parametrized by t , as depicted in the Figure – the $A(t)$ piece being a Dirichlet condition, the $B(t)$ piece a Neumann condition. The strategy is then to find $\Omega(z)$ such that $\text{Re } (\Omega(z))$ satisfies the boundary conditions.

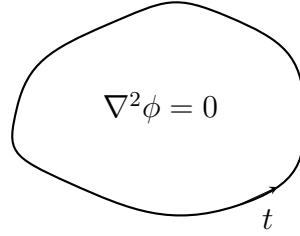


Figure 10.1: The complex Laplace problem: $\nabla^2 \phi = 0$ inside the domain, with $A(t)\phi + B(t)\partial\phi/\partial n = C(t)$ on the boundary, parametrized by t .

10.2.1 Example: the semi-infinite rectangle revisited

Consider the semi-infinite rectangle shown in the Figure: $\phi = 0$ on the two sides Γ_1 ($z = iy, y > 0$) and Γ_2 ($z = b + iy, y > 0$), and $\phi = \phi_0$ on the bottom Γ_3 ($z = x, 0 \leq x \leq b$). Consider the function

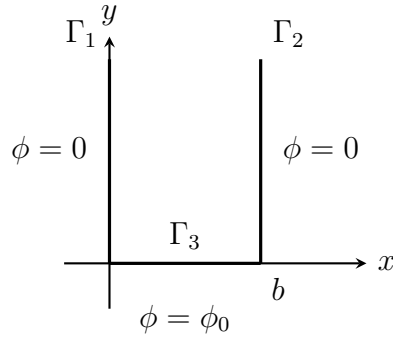


Figure 10.2: The semi-infinite rectangle: $\phi = 0$ on Γ_1 ($z = iy, y > 0$) and Γ_2 ($z = b + iy, y > 0$), while $\phi = \phi_0$ on the bottom Γ_3 ($z = x, 0 \leq x \leq b$).

$$e^{iz\pi/b} \quad (10.13)$$

It is real on Γ_1 and Γ_2 , and a pure phase on Γ_3 . Writing

$$\Omega \propto i \ln e^{iz\pi/b} \quad (10.14)$$

gives $\text{Re } \Omega = u = 0$ on Γ_1 and Γ_2 , since

$$\ln \xi = \ln |\xi| + i \arg(\xi) \quad (10.15)$$

But we also want $\Omega(z) = \text{constant}$ on Γ_3 . How to turn a pure phase into a constant? Note that

$$\frac{1 + e^{i\alpha}}{1 - e^{i\alpha}} \text{ is imaginary: } \frac{1 + e^{i\alpha}}{1 - e^{i\alpha}} \frac{1 - e^{-i\alpha}}{1 - e^{-i\alpha}} = \frac{2i \sin \alpha}{2 - 2 \cos \alpha} \quad (10.16)$$

Hence we take instead

$$\Omega(z) \propto i \ln \frac{1 + e^{iz\pi/b}}{1 - e^{iz\pi/b}} \quad (10.17)$$

and, fixing the normalization so that $\phi = \phi_0$ on Γ_3 ,

$$\Omega(z) = -i \frac{2}{\pi} \phi_0 \ln \frac{1 + e^{iz\pi/b}}{1 - e^{iz\pi/b}} \quad (10.18)$$

Then $\text{Re } \Omega = \phi$ solves the problem – this is the example from before!

10.3 Conformal maps

More generally, if $\Omega(z)$ is a solution to a complex Laplace problem in a domain D , and f is a map

$$z = f(\xi) \quad (10.19)$$

taking a domain D' to D , then

$$\chi(\xi) \equiv \Omega(f(\xi)) \quad (10.20)$$

is a solution to the complex Laplace problem on D' – the boundary conditions carry over. Particularly powerful maps are *conformal transformations*: they preserve angles and can be constructed systematically.

10.3.1 Example: a conducting corner

Consider two conducting half-planes meeting at a right angle, as shown in the Figure: Γ_1 along $\arg z = 0$ held at $\phi = \phi_0$, and Γ_2 along $\arg z = 3\pi/2$ held at $\phi = \phi_0 + K[?]$, with D the wedge $0 \leq \arg z \leq 3\pi/2$ between them. Use the map

$$\xi = \frac{2}{3\pi} \ln z \quad (10.21)$$

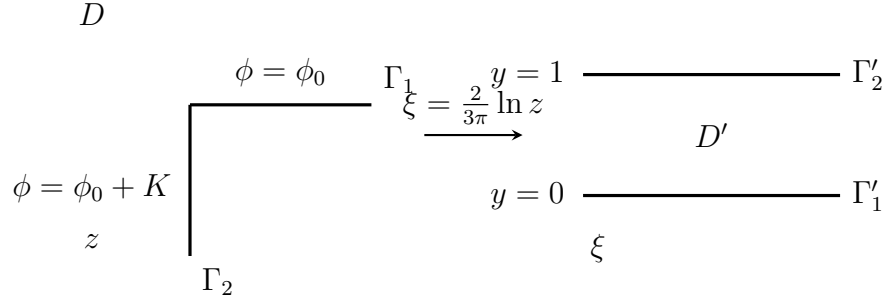


Figure 10.3: A conducting corner in the z -plane: Γ_1 ($\arg z = 0$) at $\phi = \phi_0$ and Γ_2 ($\arg z = 3\pi/2$) at $\phi = \phi_0 + K$; the map $\xi = (2/3\pi) \ln z$ takes the wedge D to the strip D' between $y = 0$ and $y = 1$ in the ξ -plane.

which takes D to the strip D' between $y = 0$ (Γ'_1) and $y = 1$ (Γ'_2), where y now denotes the imaginary part of ξ . In the strip, the potential depends only on y , and

$$\frac{\partial^2 \phi}{\partial y^2} = 0 \Rightarrow \phi = K y + \phi_0 \quad (10.22)$$

Complexifying,

$$\chi(\xi) = -i K \xi + \phi_0 \quad (10.23)$$

Mapping back to the z -plane,

$$\Omega(z) = -\frac{2iK}{3\pi} \ln z + \phi_0 \quad (10.24)$$

so that

$$\phi = \operatorname{Re}(\Omega(z)) = \frac{2K\theta}{3\pi} + \phi_0 \quad (10.25)$$

with $\theta = \arg z$ – the electric field lines are circles about the corner.

10.3.2 Example: Green's function on a disk

Consider a disk-shaped domain D with a point charge q inside, with $\phi = 0$ on the lower boundary arc Γ_1 and $\partial\phi/\partial n = 0$ on the upper arc Γ_2 , as depicted in the Figure. We need the Green's function on $\mathbb{C}$: from before,

$$G(\vec{x}, \vec{x}') = \ln |\vec{x} - \vec{x}'| \quad (10.26)$$

which complexifies to

$$g(z, z') = \ln(z - z') \quad , \quad \text{Re}(g) = G \quad (10.27)$$

Use the conformal map

$$z = \frac{t - i}{t + i} \Rightarrow t = -i \frac{z + 1}{z - 1} \quad (10.28)$$

which takes the disk to the upper half t -plane D' : Γ_1 maps to the positive real axis Γ'_1 , Γ_2 to the negative real axis Γ'_2 , and the charge – placed at the center $z = 0$ – to $t = i$. Apply another map

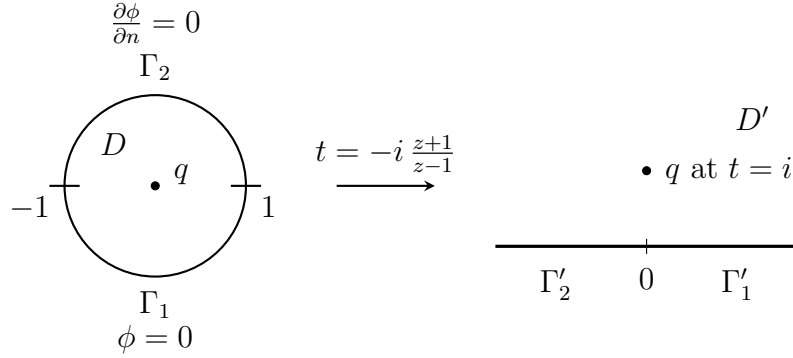


Figure 10.4: A disk with mixed boundary conditions – $\phi = 0$ on Γ_1 , $\partial\phi/\partial n = 0$ on Γ_2 , point charge q inside – mapped to the upper half t -plane; the tick marks at $z = \pm 1$, where the boundary condition changes type, map to $t = \infty$ and $t = 0$.

$$\xi = t^{1/2} \quad (10.29)$$

which takes the upper half plane to the quarter plane: Γ''_1 is the positive real axis, where $u = 0$, while Γ''_2 is the positive imaginary axis, which must be a field line, $v = 0$; the charge sits at

$$\xi_0 = e^{i\pi/4} \quad (10.30)$$

Now we can use the method of images with 3 images, as in the Figure – an image $+q$ at $-\xi_0^*$, mirroring the charge across Γ''_2 with the same sign, and images $-q$ at ξ_0^* and $-\xi_0$ with opposite sign across Γ''_1 :

$$\chi(\xi) = q \ln \frac{(\xi - \xi_0)(\xi + \xi_0^*)}{(\xi - \xi_0^*)(\xi + \xi_0)} \quad (10.31)$$

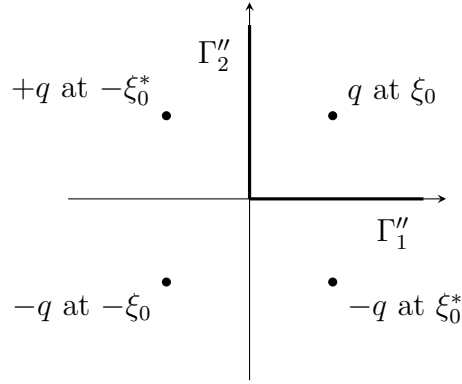


Figure 10.5: The quarter plane in the ξ -plane: the charge q at $\xi_0 = e^{i\pi/4}$ and its three images – $+q$ at $-\xi_0^*$, $-q$ at ξ_0^* , $-q$ at $-\xi_0$ – enforcing $u = 0$ on Γ''_1 and $v = 0$ on Γ''_2 .

Mapping back through (10.20), with $\xi = t^{1/2}$ and $t = -i(z+1)/(z-1)$,

$$\Omega(z) = \frac{q}{2} \ln \frac{\left(\sqrt{\frac{z+1}{z-1}} + 1\right) \left(\sqrt{\frac{z+1}{z-1}} - i\right)}{\left(\sqrt{\frac{z+1}{z-1}} - 1\right) \left(\sqrt{\frac{z+1}{z-1}} + i\right)} \quad (10.32)$$